INDIA VS NEW ZEALAND

T20 World Cup final Prediction

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Project Overview

Exploring Match Prediction Techniques

Objective

The primary objective of this project is to analyze historical match data and develop a model to accurately predict match outcomes using logistic regression techniques.

Significance

This project aims to enhance sports analytics by providing insights into match probabilities, thereby supporting teams and enthusiasts in understanding potential outcomes and strategic decisions in cricket.

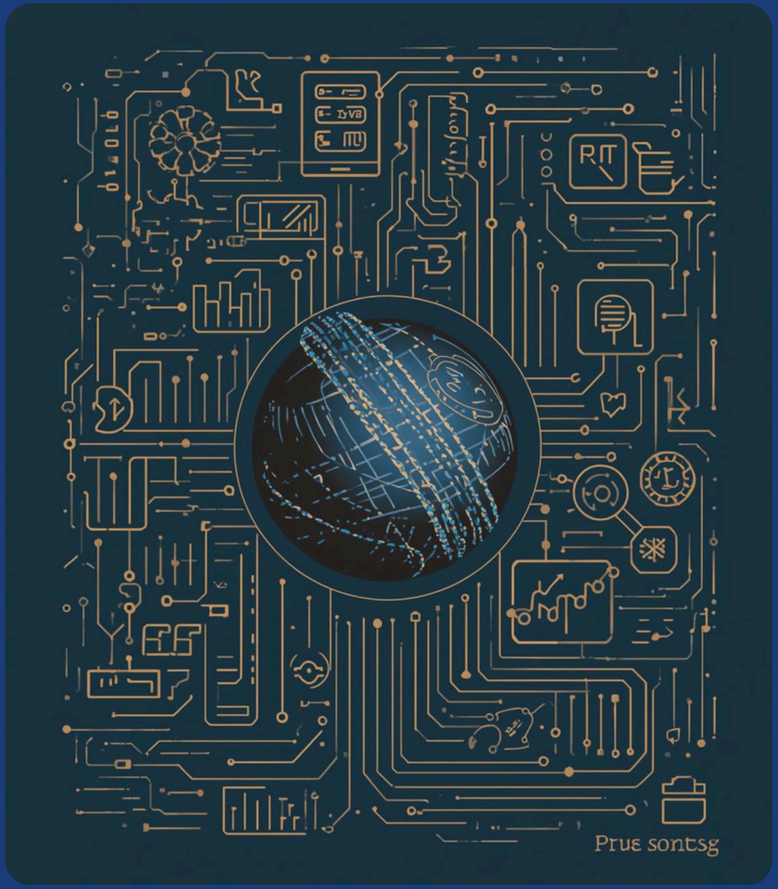
Dataset Overview and Preparation

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
df=pd.read_csv('cricket.csv')
df.head()
```

	Match_ID	Date	Venue	Team_A	Team_B	Stage	Team_A_Ranking	Team_B_Ranking	Team_A_Form	Team_B_Form	...	HeadToHead_B_Wins	Venue_HomeAd
0	M0001	2026-03-06	Chennai	New Zealand	Scotland	Super8	5	13	76.97	54.62	...	4	
1	M0002	2026-03-07	Mumbai	South Africa	Sri Lanka	Final	6	9	64.19	58.75	...	6	
2	M0003	2026-03-02	Kolkata	India	PNG	Group	1	20	69.87	45.14	...	0	
3	M0004	2026-02-22	Colombo	Netherlands	Ireland	Group	11	12	85.59	61.94	...	4	
4	M0005	2026-02-28	Mumbai	Namibia	South Africa	Group	15	6	45.98	88.52	...	15	

5 rows × 21 columns



Logistic Regression Model Overview

```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

X = df_vif.drop("Winner",axis=1)
y = df_vif["Winner"]

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)

model = LogisticRegression(max_iter=1000)

model.fit(X_train,y_train)
```

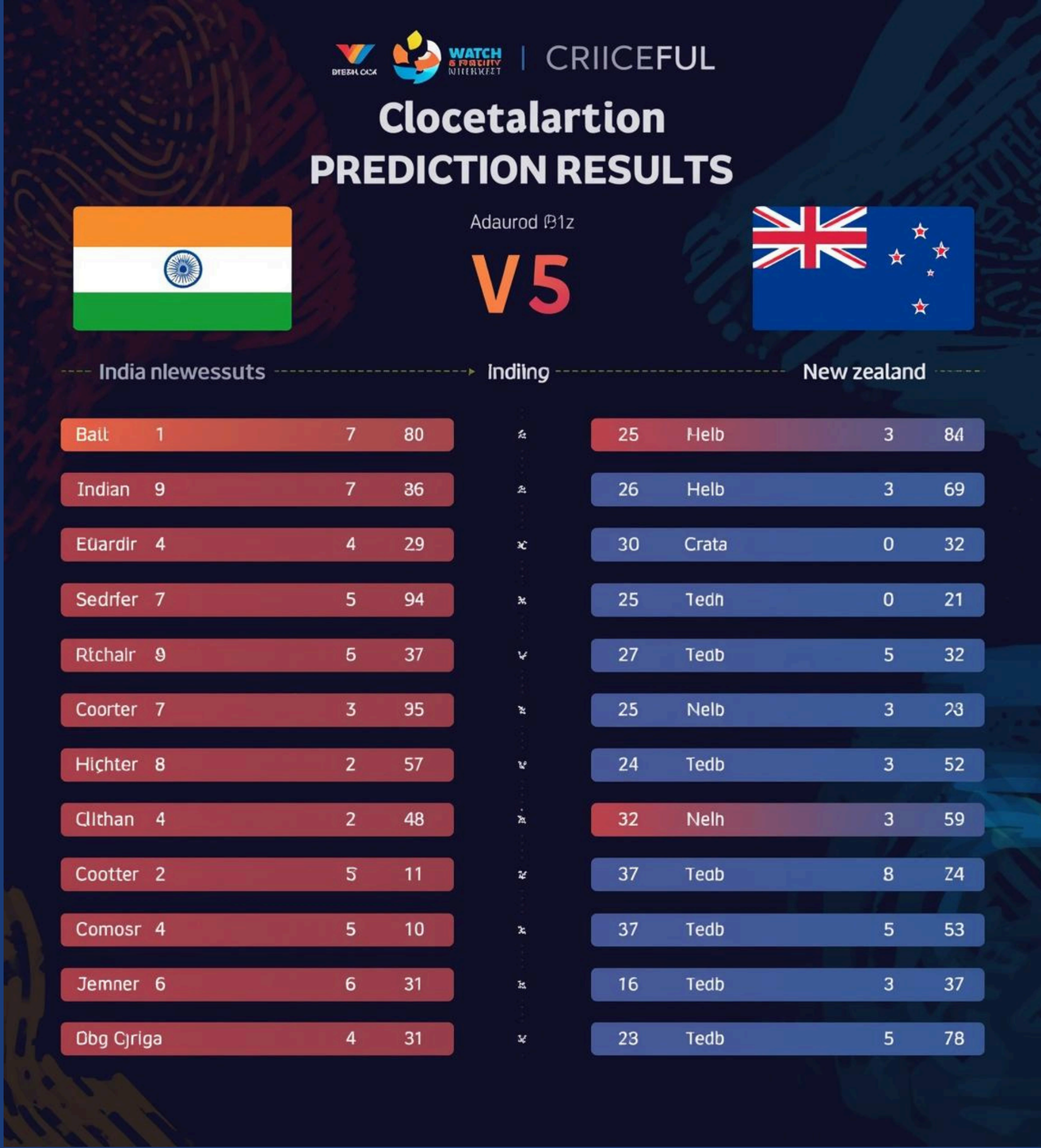
```
LogisticRegression
▶ Parameters
```

The logistic regression model utilizes historical match data to predict outcomes, focusing on key performance indicators. This model offers valuable insights into match dynamics and win probabilities for India versus New Zealand.

58.76%

India Win Probability for Match

The analysis indicates a **high probability** of India winning the match against New Zealand, demonstrating the effectiveness of the logistic regression model in predicting match outcomes.



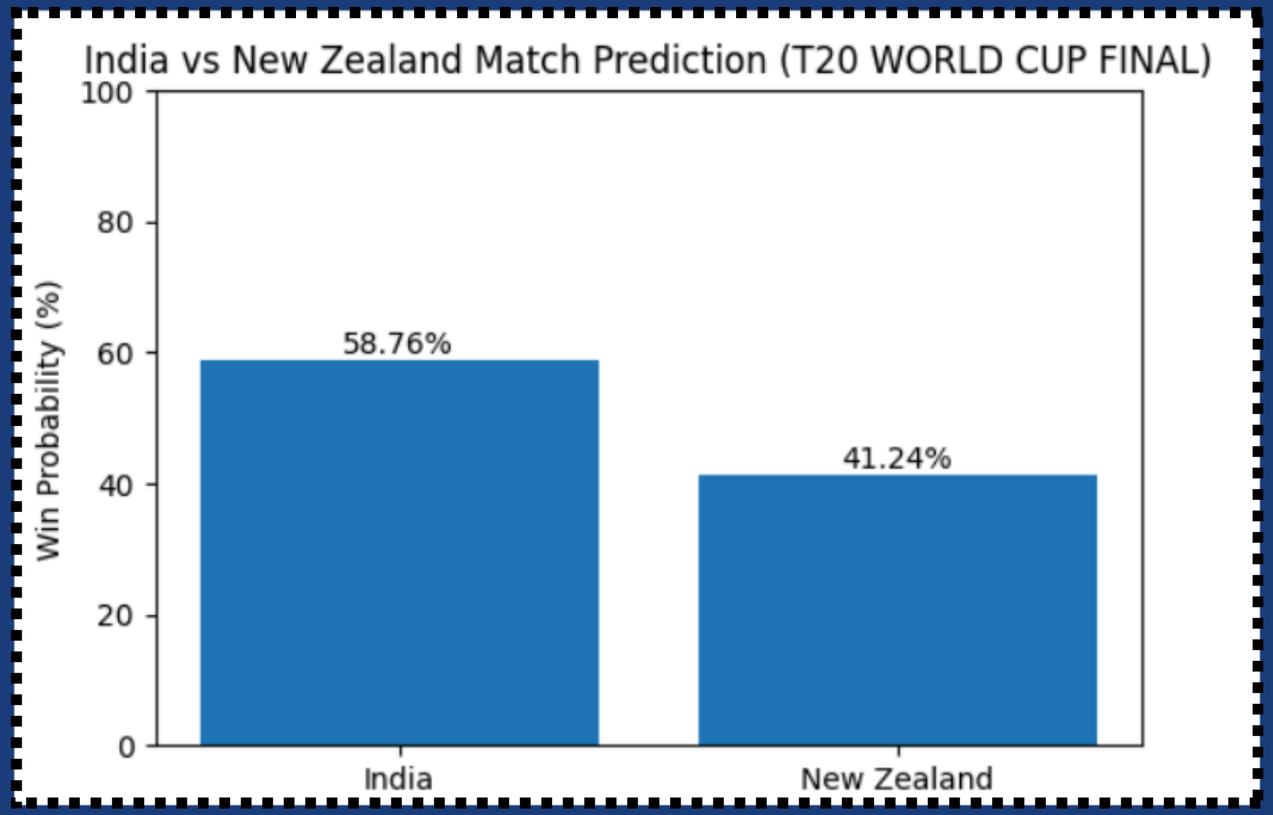
Win Probability Visualization

```python
india_prob = ind_nz["Team_A_Win_Prob"].mean()
nz_prob = ind_nz["Team_B_Win_Prob"].mean()

print("Final Match Prediction")
print("----------------------")
print(f"India Win Probability : {india_prob*100:.2f}%")
print(f"New Zealand Win Probability : {nz_prob*100:.2f}%")

if india_prob > nz_prob:
    print("Predicted Winner : India")
else:
    print("Predicted Winner : New Zealand")
```

```
Final Match Prediction
----------------------
India Win Probability : 58.76%
New Zealand Win Probability : 41.24%
Predicted Winner : India
```

This chart illustrates the predicted win probabilities for India and New Zealand in the T20 World Cup match. India's probability stands at **58.76%**, while New Zealand's is **41.24%**.

Conclusion



A successful prediction journey